

MATLAB Radiotherapy Utilities – Function Reference (with Previous Names)

This document summarizes all MATLAB functions developed, grouped by category. Each function includes:

- **Previous Name** – original function name before shortening
- **Current Name** – standardized short matRad name
- **Description** – what it does
- **Call** – example usage
- **Status** – development/stability indicator

✔ **Legend:** ● Stable — tested and ready for integration ● Needs refinement — partially implemented or pending testing ● Experimental — draft or conceptual ✖ Deprecated — no longer in use

2. Data Import / Export

Previous Name	Current Name	Description	Call
–	matRad_dataImpWrap	High-level wrapper: for given patient, path, and plan, load from MAT if present, otherwise import from DICOM using matRad_dicomImpDose	[ct, cst, doseCube, resultGUI] = matRad_dataImpWrap(patientName, inputPath, planName, 'fallbackPath', fbPath, 'saveDir', outDir)
–	matRad_dicomImpDose	Import CT, RTSTRUCT and one or multiple RTDOSE series; saves matRadPatient_<pat>_<plan>.mat or split ct_cst_<pat>.mat + doseCube_<pat>_<plan>.mat	matRad_dicomImpDose(patientName, dicomPath, {'PlanA', 'PlanB'}, 'fallbackPath', path, 'saveDir', outDir)
–	matRad_matLoad	Load CT, CST, doseCube, resultGUI by patient/plan from MAT (exact or wildcard search, supports split CT/CST + dose files)	[ct, cst, doseCube, resultGUI] = matRad_matLoad(patientName, planName, 'searchDir', folder)
-	matRad_matLoadDirect	Load variables selectively from MAT file by path	[ct, cst, pln, stf, resultGUI] = matRad_matLoadDirect(filePath)
–	matRad_findPatientMats	Find all MAT files for a patient, optionally filtered by plan/beam string	matFiles = matRad_findPatientMats('SP03', 'PassivePlusArc', 'searchDir', dir)
Workspace Based			
matRad_filenameGenerate	matRad_wsFileGen	Generate standardized workspace filenames from plan gantry spacing and dose grid resolution (e.g. Base_plnstfdijGantry10Res222_2025-01-05_1430.mat)	[filePath, fileFolder, fileName] = matRad_fileGen(fileFolder, pln, workspaceType)
matRad_fileGet	matRad_wsFileFind	Find simulation/workspace MAT files by gantry spacing and dose grid resolution (e.g. Base_plnstfdijGantry10Res222_*.mat)	[filePath, fileFolder, fileName, allMatches] = matRad_fileGet(folder, gantrySep, [dx dy dz], workspaceType)
matRad_filePatientPath	matRad_patPath	Locate patient folders/files recursively	[targetPath, patientFolder] = matRad_patPath(patientID, searchFolder, searchFile)
matRad_savePatientMat	matRad_savePatientMat	Save patient data to MAT	matRad_savePatientMat(patientName, ct, cst, pln, stf, resultGUI, saveDir, saveName)

Key rule:

Patient / clinical plan IO → always use:
 matRad_dicomImpDose (save using patient+plan)
 matRad_matLoad / matRad_findPatientMats
 matRad_dataImpWrap
 Simulation / optimization workspace IO → use:
 matRad_fileGen (or matRad_wsFileGen)
 matRad_fileGet (or matRad_wsFileFind)
 matRad_matLoadDirect to load from a chosen workspace file

3. Structure and Contour Tools (CST/VOI)

Previous Name	Current Name	Description	Call	Status
matRad_cstAddDoseObjectives	matRad_cstAddObj	Add dose objectives manually	cst = matRad_cstAddObj(cst, VOIs, basePenalty)	●
matRad_cstAddDoseObjectivesMerged	matRad_cstAddObjMged	Merge objectives from multiple sources	cst = matRad_cstAddObjMged(cst, [3 5], dvhRef, 100, 'sourceType', 'reference')	●
matRad_cstEmulateDVHObjectives	matRad_cstAddObjRefDVH	Generate DVH-based objectives	cst = matRad_cstAddObjRefDVH(cst, VOIs, dvhBase, basePenalty, metrics)	●
matRad_cstExtractDoseObjectives	matRad_cstExtDoseObj	Extract all dose objectives from CST	[metricStrings, objStruct] = matRad_cstExtDoseObj(cst, VOIs)	●
matRad_cstMetrictoStruct	matRad_cstMetric2Struct	Convert human-readable metric to CST object	obj = matRad_cstMetric2Struct(metricStr)	●
matRad_cstPrint	matRad_cstPrint	Print objectives	matRad_cstPrint(cst)	●
matRad_setupCST	matRad_cstSetup	Initialize CST	[cst, presDose, ixPTV, ixExt] = matRad_cstSetup(cst, VOINames, VOISites, mode)	●
matRad_VOIcompare	matRad_VOIcmp	Compare VOIs between plans	[report, relDiffs] = matRad_VOIcmp(cst, qi, qiRef, {'D_2', 'mean'})	●
matRad_VOICreateRings	matRad_VOICreateRings	Create ring structures	cstOut = matRad_VOICreateRings(ct, cst, ring_mm, ixRefVOI, marginPTVRing_mm, visualize)	●
matRad_VOIDoseThreshold	matRad_VOIDoseThr	Filter VOIs by dose threshold	cstFilt = matRad_VOIDoseThr(cst, doseCube, 10, 'above', 'voxels.txt')	●
matRad_VOIDoseThresholdMask	matRad_VOIDoseThrMask	Mask voxels above threshold	mask = matRad_VOIDoseThrMask(cst, doseCube, {'L Lung'}, 10)	●
matRad_VOIfilterDoseThreshold	matRad_VOIFilterDoseThr	Legacy function for thresholding	cstFilt = matRad_VOIFilterDoseThr(cst, doseCube, 10, 'below')	●
matRad_VOIfindIndex	matRad_VOIFindIx	Find VOI index	ixVOI = matRad_VOIFindIx(cst, aliases)	●

matRad_VOIIrradiatedHealthyTissue	matRad_VOIHealthy	Define healthy tissue VOIs above threshold	[mask, cst, maskThr] = matRad_VOIHealthy(cst, doseCube, 10, false)	●
N/A	matRad_VOIHealthy	Create VOIs for irradiated healthy tissue with island-size filtering	[healthyMask, cstNew, healthyAboveThrMask] = matRad_VOIHealthy(cst, doseCube, doseThreshold, includeTargets, minIslandVox)	●
matRad_VOIOperations	matRad_VOIOps	Combine VOIs via set operations	[cst, newIx] = matRad_VOIOps(cst, ix1, ix2, operation, newName)	●
matRad_OARFindIx	matRad_VOIOARFindIx	Finds all indexes of OARs	ixOARs = matRad_VOIOARFindIx(cst)	●
	matRad_getVoiMask	Returns a binary mask for a VOI	mask = matRad_getVoiMask(ct, cst, voiName, slice)	●
	matRad_fixExternalMask	Repair missing voxels or gaps in External contour	[maskExternalFixed, cstOut, alteredSlices] = matRad_fixExternalMask(ct, cst, HUrang, sliceList)	●

4. Dose Objectives and Constraints

Previous Name	Current Name	Description	Call	Status
matRad_cstAddDoseObjectives	matRad_cstAddObj	Add manual dose objectives to CST	cst = matRad_cstAddObj(cst, [3 5], {'D_98 minDVH 30Gy 100', 'mean 200'}, true)	●
matRad_cstAddDoseObjectivesMerged	matRad_cstAddObjMged	Merge objectives from manual, reference, or QI sources	cst = matRad_cstAddObjMged(cst, [3 5], dvhRef, 100, 'sourceType', 'reference')	●
matRad_cstEmulateDVHObjectives	matRad_cstAddObjRefDVH	Generate dose objectives from reference DVHs	cst = matRad_cstAddObjRefDVH(cst, VOIs, dvhBase, basePenalty, metrics)	●
matRad_cstExtractDoseObjectives	matRad_cstExtDoseObj	Extract all dose objectives from CST into readable metric strings	[metricStrings, objStruct] = matRad_cstExtDoseObj(cst, VOIs)	●
matRad_cstMetrictoStruct	matRad_cstMetric2Struct	Convert human-readable metric	obj = matRad_cstMetric2Struct(metricStr)	●

string into
dose
objective
struct

5. Beam Geometry / Energy Layers

Previous Name	Current Name	Description	Call	Status
matRad_generatePln	matRad_genPln	Create basic matRad plan structure	pln = matRad_genPln(cst, ct, gantryAngles, doseGridResolution, modality)	●
matRad_generateStfOneEL	matRad_genStf1EL	Generate single-layer STF for a beam	[stf_oneEL, stf_mb, stf_sb] = matRad_genStf1EL(ct, cst, pln)	●
matRad_generateStfSSingle...	matRad_genStfELB2	Generate STF with single EL per beam	[stf1EL, mb, sb] = matRad_genStfELB(ct, cst, pln)	●
matRad_getMachineE...	matRad_machineColorMap	Build machine-specific EL → RGB mapping for plotting	cmap = matRad_machineColorMap(stf)	●
matRad_showSpotsInSlice	matRad_showSpots	Overlay all proton spot positions on dose slice	matRad_showSpots(ct, cst, stf, doseCube, weights)	●

6. Dose Calculation / Comparison

Previous Name	Current Name	Description	Call
matRad_calcQIndAdapted	matRad_calcQI	Compute DVH-based quality indicators	qi = matRad_calcQI(cst, pln, doseCube)
matRad_compareTwoDoses	matRad_cmp2Doses	Compare reference vs test dose cubes; DVH overlay, slice differences, CI metrics	matRad_cmp2Doses(doseRef, doseTest, ct, cst)
	matRad_compareTwoDoses	Compare two dose distributions visually	[figCenter, figMaxDiff, figDVH] = matRad_compareTwoDoses(doseRef, doseTest, c
matRad_compareDoseDistribution	matRad_cmpDose	Comparison of two dose cubes visually	[~, ~, hfig] = matRad_cmpDose(cube1, cube2,
matRad_doseCubeExtract	matRad_doseExtract	Extract physical or RBEx dose cube from resultGUI	doseCube = matRad_doseExtract(resultGUI)
matRad_dosePerEnergyLayer	matRad_dosePerEL	Compute dose cubes per EL	doseCubes = matRad_dosePerEL(ct, cst, stf, dij, resultGUI, t
matRad_calcDoseCubePerEL	matRad_ELDoseCube	Compute dose cubes per energy layer using spot masks	eLayerStruct = matRad_ELDoseCube(dij, weigh eLayerStruct)
matRad_linearizeDose	matRad_linearDose	Flatten DIJ into sparse form, compute dose per voxel, reshape to 3D, optional plotting	dpDose_linear = matRad_linearDose(dij, w, p cst, 1)
	matRad_cutDoseAxial	Set dose to zero on selected axial slices	doseOut = matRad_cutDoseAxial(doseIn, ct, n zRef2)
	matRad_cutDosePipeline	Cut specified axial slices to zero and visualize with matRad_showAllPlanes	[doseCut, hFig] = matRad_cutDosePipeline(dc cst, mode, zRef, zRef2, nSlicesTarget, plan

7. Plotting, DVH and Visualization Tools

Previous Name	Current Name	Description	Call	Status
matRad_plotCTSlice	matRad_plotCT	Plot CT slice	matRad_plotCT(ct, slice)	●
matRad_plotEnergyLayerHistogram	matRad_plotELhist	Plot number of spots per EL	matRad_plotELhist(ax, stf)	●
matRad_plotSpotsSlice	matRad_plotSpotsSlice	Plot all spots in a CT slice	matRad_plotSpotsSlice(ax, ct, stf, markerSize, weights, showRayTracing, useGeoSpots, sliceIdx)	●
matRad_showDVHAdapted	matRad_showDVH	Show adapted DVH	matRad_showDVH(cst, doseCube)	●
matRad_showMultiDVH	matRad_showMultiDVH	Plot multiple DVHs from multiple sources	matRad_showMultiDVH(dvhResults, cst)	●
matRad_showSliceAndDVH	matRad_showSliceDVH	Display dose slice and DVH side-by-side	[hleg, dvh] = matRad_showSliceDVH(ct, cst, doseCube)	●
matRad_showSliceFast	matRad_showSliceFast	Quick slice visualization	[slice, hleg] = matRad_showSliceF(ct, cst, doseCube, slice, doseWindow)	●
	matRad_showAllPlanes	Visualize CT, CST, and optionally dose in 3 planes and highlight altered slices if provided	hFig = matRad_showAllPlanes(ct, cst, doseCube, nSlicesTarget, alteredSlices, plane)	●
	matRad_saveAllPlanes	Save all plane figures from matRad_showAllPlanes to folder	matRad_saveAllPlanes(hFig, exportFolder, resolution)	●

8. Spot and Energy Layer Analysis

Previous Name	Current Name	Description	Call
matRad_energyLayer_attachRays...	matRad_attachRays	Attach spot metadata to EL structure	e1Struct = matRad_attachRays(e1Struct, stf)
matRad_checkEnergies	matRad_chkEnergy	List unique energy layers per beam, counts and total rays	[allE, nE, nR] = matRad_chkEnergy(stf)
matRad_dosePerEnergyLayer	matRad_dosePerEL	Compute dose cubes per EL	doseCubes = matRad_dosePerEL(ct, cst, stf, dij, resultGUI, threshold%)
matRad_energyLayer_perVOIDose	matRad_dosePerVOI	Compute per-VOI dose per EL	[voiDoseByEL, qiByEL] = matRad_dosePerVOI(cst, allELStruct, dij, w, refGy, refVol)
matRad_EnergyLayerAnalysis	matRad_ELAnalysis	High-level pipeline: weight layers, compute per-EL dose, attach rays, per-VOI metrics, visualize	[voiDoseByEL, qiByEL, topEL] = matRad_ELAnalysis(cst, stf, dij, ct, resultGUI, ixTarget, ixOAR)

		Simplify STF	
matRad_generateStfSSingleEn...	matRad_genStfELB	to single EL per spot	[stf1EL, mb, sb] = matRad_genStfELB(ct, cst, pln)
matRad_ELayerMergeSimilar matRad_mergeEL Merge adjacent/nearby energy layers into representative center layers merged = matRad_mergeEL(topEL, mergeRange, mergeTo1) ● matRad_energyLayer_plotDosePerEL matRad_plotDoseEL Grid figure of per-EL doses with spots overlay t = matRad_plotDoseEL(ct, cst, stf, topELStruct, qiByEL, targetName) ● matRad_plotEnergyLayerPerWeight matRad_plotELW Plot per-weight energy layer representation matRad_plotELwt(ax, stf) ● matRad_plotEnergyLayerH... matRad_plotELhist Plot number of spots per EL matRad_plotELhist(ax, stf) ● matRad_plotE...CAndW matRad_plotELhistCW Plot histogram of counts and weights per EL matRad_plotELhistCW(ax, stf) ● matRad_plotE...CAndWStacked matRad_plotELhistCWStack Plot stacked histogram of counts and weights per EL matRad_plotELhistStack(ax, stf) ● matRad_plotE...StackedW matRad_plotELhistWStack Plot stacked histogram of weights per EL matRad_plotELhistSwt(ax, stf) ● matRad_plotE...H matRad_plotELwtHist Plot total spot weights per EL matRad_plotELwtHist(ax, stf, 'perBeam') ● matRad_plotGeoSpot matRad_plotGeoSpot Plot spot geometry visualization matRad_plotGeo(ct, stf, iBeam) ● matRad_plotSingleRay matRad_plotRay Visualize spots from specific ray index medianSpotCube = matRad_plotRay(ax, stf, iRayTarget, markerSize, weights, useGeoSpots) ● matRad_removeSpotsOverlappingVOIs matRad_rmSpotsVOI Remove spots overlapping VOIs; return retained mask & indices [logicalMask, removedIdx] = matRad_rmSpotsVOI(stf, ct, cst, ixVOI, iBeam) ● matRad_spotix matRad_spotix Compute global 1D index of a spot ix = matRad_spotIx(stf, iBeam, iRay, iSpot) ● matRad_calcSpotW... matRad_spotWeight Normalizes and assigns spot weights, and constructs a 3D weight matrix [stf, wMatrix] = matRad_calcSpotWeightsAndMatrix(stf, weights) ● matRad_analyzeSpotsPerBeam matRad_spotsAnalyze ERROR Visual/interactive analysis of one beam [doseHeavy, stats] = matRad_spotsAnalyze(iBeam, ct, cst, stf, dij, resultGUI, mode, threshold) ● matRad_computeSpotPositions_Geo matRad_spotsPosGeo Computes spot positions based on beam setup geometry stf = matRad_spotsPosSiddon(ct, stf, machine) ● matRad_computeSpotPositions_Siddon matRad_spotsPosSiddon Compute spot positions using Siddon algorithm stf = matRad_spotsPosSiddon(ct, stf, machine) ● matRad_computeSpotStats matRad_spotsStats Compute summary spot statistics stats = matRad_spotsStats(stf, w) ● matRad_analyzeSpotsPerBeamStats matRad_spotsStatsBeam Numeric stats per beam for filtered spots [doseHeavy, stats, wHeavy, mask] = matRad_spotsStatsBeam(iBeam, ct, cst, stf, dij, resultGUI, mode, threshold) ● matRad_weightEnergyLayers matRad_weightEL Summarize weights per EL, compute dose contribution, identify top layers [tSummary, top, all, locs] = matRad_wtEL(cst, stf, dij, w, oarIdx, wThresh) ● matRad_getTopELStruct matRad_getTopELStruct Subset top energy layers struct based on table topELStruct_red = matRad_getTopELStruct(voiDoseByEL, topELStruct, N) ● matRad_getSingleEL matRad_getSingleEL Extract single EL dose + metrics structs [elDoseStruct, elQiStruct] = matRad_getSingleEL(topELStruct, qiByEL, elName) ● matRad_optSingleEL matRad_optSingleEL Reoptimize a single EL via spot removal [doseELOpt, wELOpt, weightsFiltered] = matRad_optSingleEL(dij, cst, pln, wAll, elStruct) ● matRad_getMachineE... matRad_machineColorMap Build machine-specific EL → RGB mapping for plotting cmap = matRad_machineColorMap(stf) ●			

9. FLASH / DMF

Previous Name	Current Name	Description	Call
matRad_restoreFLASH_detectDMF	matRad_detectFLASH	Detect likely FLASH DMF-affected voxels using dose-only heuristics (threshold, valley, morph) and reconstruct pre-DMF dose cube. Returns reconstructed dose, mask, and report with statistics.	[doseFLASH, flashMask, report] = matRad_detectFLASH(doseEff, cst, ixTarget) [doseFLASH, flashMask, report] = matRad_detectFLASH(doseEff, cst, ixTarget, 'DMF', 2.0, 'DoseThr', 10, 'Method', 'valley', 'DoPlot')
matRad_evaluateFLASHBeams	matRad_evalFLASH	Evaluate multiple FLASH proton beam plans using dose metrics: PTV, FLASH target,	matRad_evalFLASH(beanSearch, cst, ptvAlias, flashoarAliases, prescribedDose)

		OARs, Conformity Index (CI), and objective values. Prints a summary table.	
matRad_FLASHApplyDMF	matRad_flashApplyDMF	Apply FLASH DMF to a 3D dose cube using pre-computed voxel masks or per-VOI DMF. Supports optional output summary.	doseCubeMod = matRad_flashApplyDMF(cst, doseCube, voiSelection, doseAboveThrMask) doseCubeMod = matRad_flashApplyDMF(cst, doseCube, voiSelection, doseAboveThrMask, outputTxtFile)
matRad_FLASHDMFPipeline	matRad_flashPipeline	Complete FLASH dose-modification pipeline. Applies a Dose Modifying Factor (DMF) to voxels above a threshold in selected OARs, healthy tissue, or both. Returns modified dose cube, binary masks, and optionally updates CST.	[doseCubeMod, maskStruct, cstOut] = matRad_flashPipeline(cst, doseCube, DMF, voiSelection, doseThreshold) [doseCubeMod, maskStruct, cstOut] = matRad_flashPipeline(cst, doseCube, DMF, voiSelection, doseThreshold, outputTxtFile, includeHealthyTissue, returnMasks, applyMode)
matRad_recoverFlashDose	matRad_recoverFlash	Reconstruct original FLASH dose distribution from DMF-modified dose cube. Generates dose difference map and FLASH-effect mask. Optional QC plotting included.	[doseFLASH, doseDiff, flashMask] = matRad_recoverFlash(ct, cst, doseEff, DMF, FLASHt1, doPlot)
matRad_recoverFlashDoseSimple	matRad_recoverFlashSimple	Threshold-based reconstruction of pre-DMF FLASH dose. Provides dose difference and voxel mask. Simple, fast	[doseFLASH, doseDiff, flashVoxels] = matRad_recoverFlashSimple(ct, cst, doseEff, DMF, FLASHthresh, doPlot)

version for QC
purposes.

Current Name	Description	Call	Status
matRad_flashVoxels	prepare voxel masks and CST entries for FLASH analysis	[flashMask, flashOARmask, cstFLASH] = matRad_flashVoxels(cst, doseCube, doseThreshold, voiSelection, includeHealthy)	
matRad_flashApplyDMF	Apply FLASH DMF to a precomputed voxel mask	doseCubeMod = matRad_flashApplyDMF(doseCube, flashMask, DMF)	
matRad_flashLog	Print FLASH preprocessing results	logFileName = matRad_flashLog(cstFLASH, DMF, doseThreshold, finalMask)	
matRad_flashPipeline	Full FLASH dose modification and visualization pipeline	[doseCubeModified, cstModified, flashMask, logFileName, hFigSlicesCenter, hFigSlicesMaxDiff, hFigDVH] = matRad_flashPipeline(cst, doseCube, doseThreshold, DMF, voiSelection, includeHealthyTissues)	

10. Miscellaneous

Previous Name	Current Name	Description	Call	Status
STAR_getPatientPaths	STAR_getPatientPaths	Helper to locate STAR patient data	[paths] = STAR_getPatientPaths(baseFolder)	
matRad_getDoseInVOI				

**** A. To be added ****